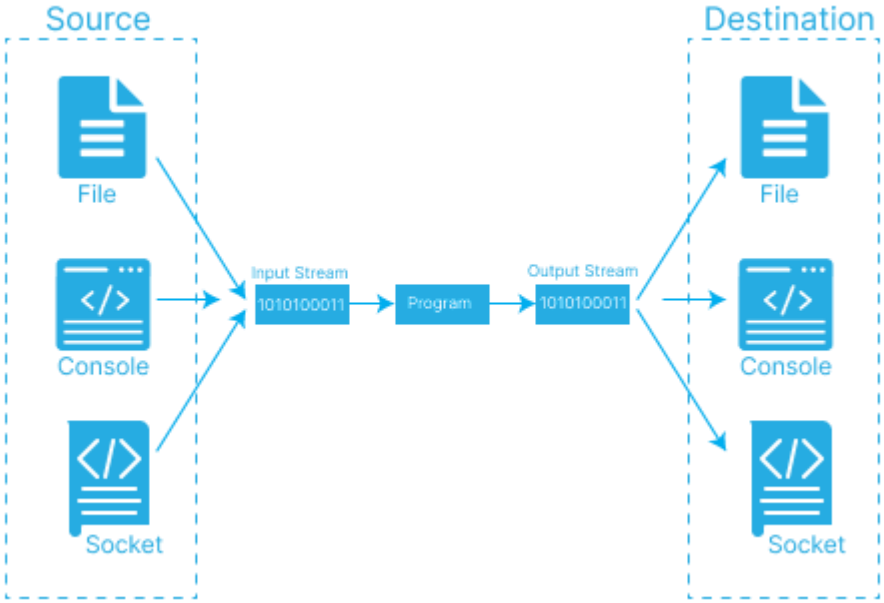
# Java Streams – Characters vs Bytes

- A Java Stream is a group of objects that can be piped together to produce the desired result.
- Streams are used in Java to transfer data between programs and I/O devices like a file, network connections, or consoles.



- In Java, character streams and byte streams are two types of input/output streams that differ mainly in how they handle data.
- Character streams are designed for handling characters (formatted text) and use Unicode encoding to represent characters. They are suitable for working with text data, such as reading/writing files that contain text.
- Byte streams are used for handling raw binary data in the form of bytes. They are suitable for reading/writing non-textual data, such as images, video files or audio files.
- Character streams automatically handle character encoding and decoding, while byte streams require explicit encoding/decoding when working with text data.
- Overall, the choice between character streams and byte streams depends on the type of data being handled, with character streams being more appropriate for text-based operations and byte streams being more versatile for low level I/O operations, such as reading and writing to network sockets.

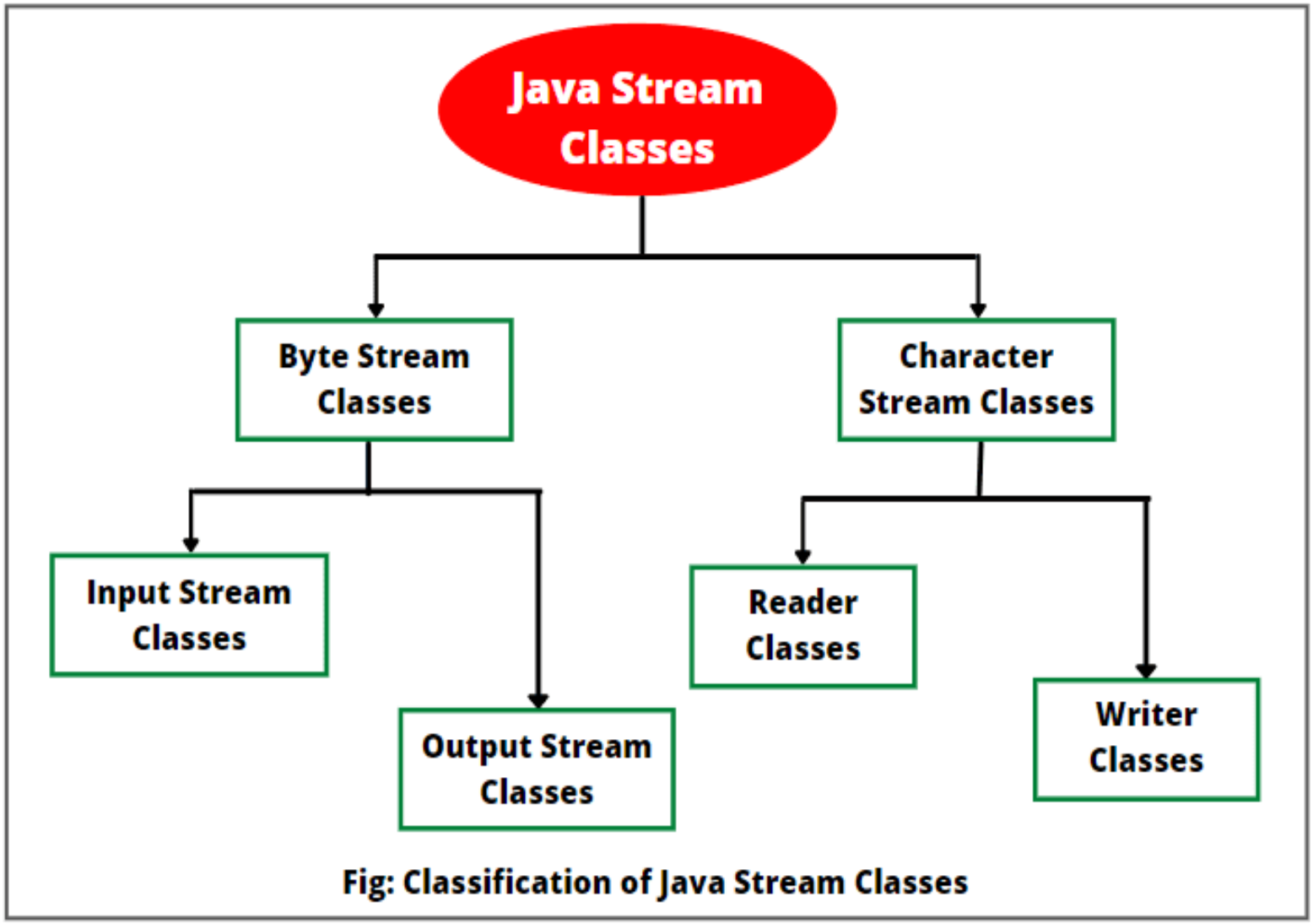
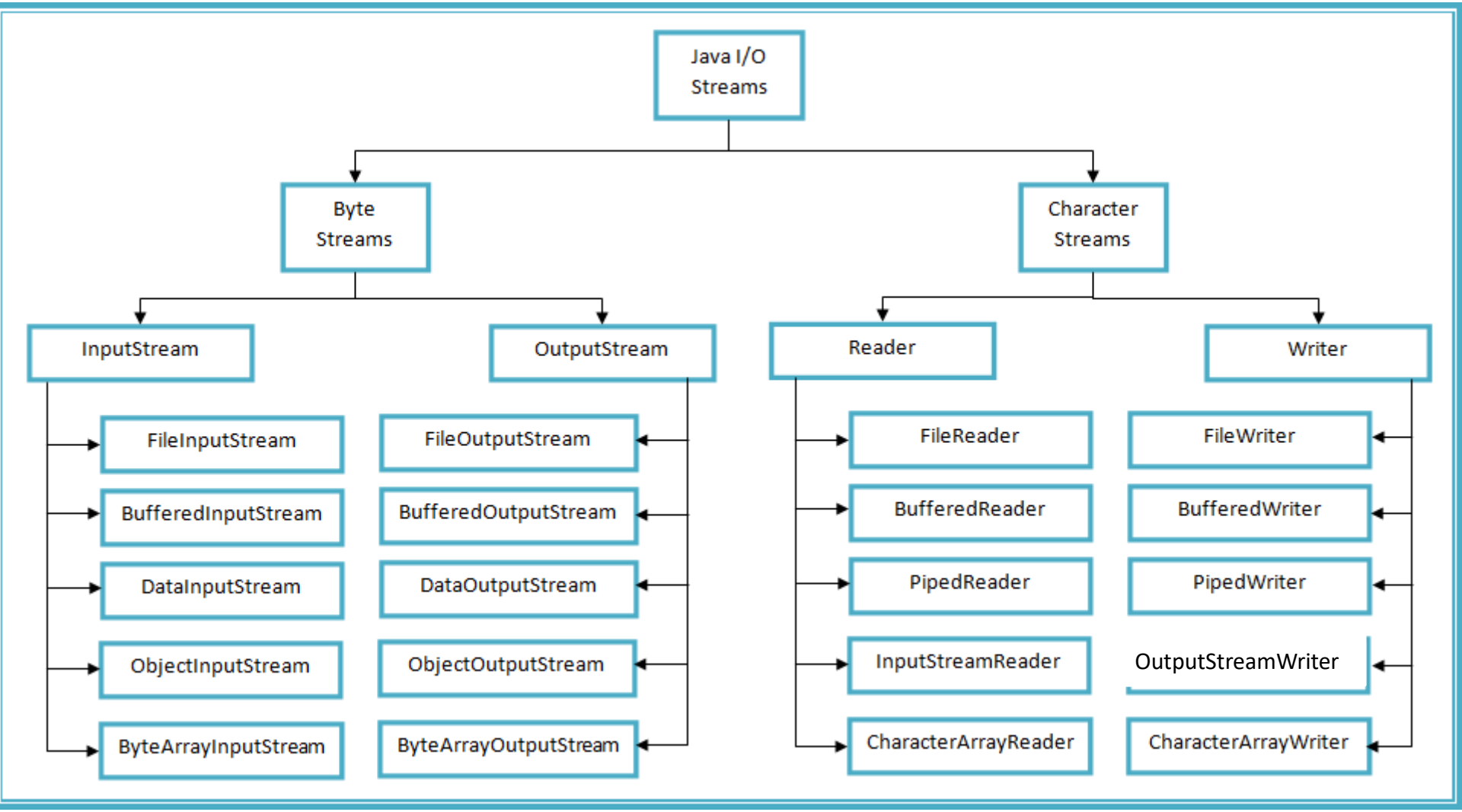



Fig: Classification of Java Stream Classes


Java I/O Streams
Byte Streams
Character Streams
InputStream
OutputStream
Reader
Writer
FileInputStream
BufferedInputStream
DataInputStream
ObjectInputStream
ByteArrayInputStream
FileOutputStream
BufferedOutputStream
DataOutputStream
ObjectOutputStream
ByteArrayOutputStream
FileReader
BufferedReader
PipedReader
InputStreamReader
CharacterArrayReader
FileWriter
BufferedWriter
PipedWriter
OutputStreamWriter
CharacterArrayWriter

## ❑ Byte Streams

- ByteStream classes read and write 8-bit data.
- Using ByteStream classes, we can save video, audio, characters, and so on.
- The java.io package contains these classes.
- The ByteStream classes are classified into two types: *InputStream* and *OutputStream*.
- These are the abstract super classes for all ***Input/Output stream classes***.

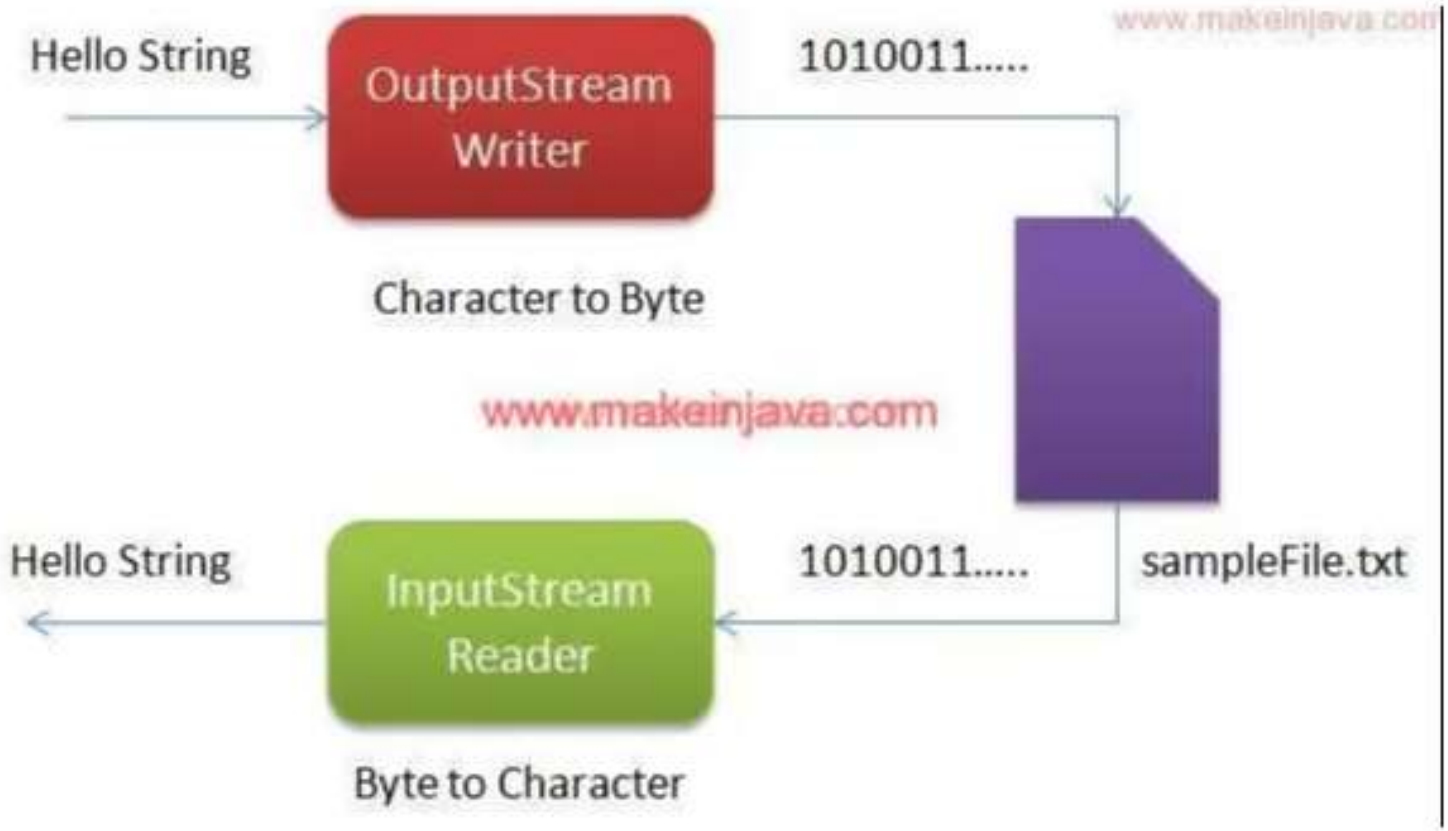
➢ ***InputStream:***

- This type of java stream reads data from a source.
- An InputStream can read data from a file, network connection, or any other I/O device.

➢ ***OutputStream*:**
The outputstream is a java stream that takes data from a Java program and sends or writes it to the destination

## ❑ Character Stream

- The Character stream is used for 16-bit Unicode input and output operations.
- Character streams would be advantageous if we wanted to copy a text file containing characters from one source to another using streams because it deals with characters.
- CharacterStream classes are used to work with Unicode characters that are 16 bits in length.
- They can work with characters, char arrays, and Strings.
- The CharacterStream classes are primarily used to read characters from the source and write them to the destination.
- CharacterStream classes are divided into two types for this purpose: ***Reader classes*** and ***Writer classes***.




OutputStreamWriter and InputStreamReader